Anomaly detection in spacecraft telemetry: an ensemble approach

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Anomaly detection in spacecraft telemetry is critical for ensuring mission safety and operational continuity. However, traditional threshold-based approaches struggle with the high-dimensional nature of real satellite data. This paper presents a hybrid ensemble framework for detecting anomalies in a sample from the European Space Agency (ESA) Spacecraft Anomaly Detection Benchmark dataset, which includes 14 years of multivariate telemetry from operational missions. Our approach wants to demonstrate robust detection across diverse anomaly manifestations while maintaining interpretability through tree-based feature importance analysis.

Keywords: Space, AI, Anomaly detection, Telemetry data analysis

1. Introduction

Spacecraft operations represent one of the most data-intensive and safety-critical domains within aerospace engineering. Continuous telemetry monitoring is essential for mission success and early fault detection. Modern satellites generate high-dimensional telemetry streams encompassing thermal sensors, power systems, attitude control, and communication subsystems, with the European Space Agency's operational missions recording up to 87 concurrent parameters sampled at regular intervals over mission lifespans spanning decades (Kotowski et al., 2023).

Traditional anomaly detection methodologies in mission control have relied predominantly on threshold-based limit checking, where individual telemetry channels are flagged when values exceed predefined operational bounds. However, the increasing complexity of spacecraft systems and the subtle, multivariate nature of real-world anomalies have exposed fundamental limitations in these rule-based approaches.

Our methodology tries to address these limitations by integrating Isolation Forest with a Telemanom-inspired predictor, utilizing XGBoost as a meta-learning layer. Through this tree-based feature importance analysis,the interpretability is increased, enabling mission operators to assess the relative contribution of structural versus temporal signals to each anomaly prediction. Evaluation is conducted emphasizing precision over recall, reflecting operational constraints where false alarms impose significant procedural and cognitive costs on mission control teams.

2. Related Work

Existing anomaly detection methodologies exhibit systematic limitations when applied to spacecraft telemetry. Statistical approaches such as global threshold methods fail to capture multivariate correlations and are highly sensitive to operational regime changes, producing excessive false alarms. Deep learning architectures exemplified by NASA's Telemanom framework, excel at modeling temporal dependencies, but come with prohibitive computational costs and lack of interpretability. Geometric outlier detectors such as Isolation Forest effectively identify structural anomalies in multivariate feature space but remain temporally insufficient, unable to detect sequence-level deviations that unfold over extended windows.

Perhaps most critically from an operational perspective, the European Space Operations Centre has identified manual threshold tuning as a significant bottleneck in current mission control workflows. Defined thresholds require continuous adjustment as missions progress through different

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phases, and the cognitive burden on flight controllers is exacerbated by high false alarm rates from poorly calibrated detectors. Seekig the ESA's Actionable AI roadmap (De Canio et al., 2023), the automation of detection boundaries and reduction of manual heuristics to enable scalable autonomous mission operations, is prioritized.

Beyond theoretical performance, the operational integration of anomaly detection algorithms is guided by strategic frameworks such as the "Actionable AI" roadmap proposed by De Canio et al. (2023). This roadmap advocates for the transition of the European Space Operations Centre toward automated mission operations by prioritizing high-confidence, interpretable outputs that minimize the cognitive load on flight controllers. To align with the A²I mandate for reducing reliance on manual heuristics, the current methodology employs a supervised meta-learning stage via XGBoost to fuse detector outputs. This approach replaces conventional, manually tuned thresholds identified by (De Canio et al., 2023) as a significant operational bottleneck with an automated decision boundary designed to enhance end-to-end reliability in mission environments.

Methodological rigor in Multivariate Time Series Anomaly Detection has recently faced scrutiny regarding the validity of common evaluation protocols. Sehili and Zhang (2024) highlight systemic flaws in algorithms that rely on Point Adjustment metrics, which may artificially inflate performance results by failing to account for true event-level detection. In accordance with this critique, the present work privileges the `Precision` metric and incorporates post-processing to ensure detections are compact and operationally relevant.

To address the dichotomy between sequence-based predictive models and geometric outlier detectors, we prepared an ensemble method. By combining temporal error signals from predictive architectures, such as Telemanom, with structural rarity cues from Isolation Forest, our ensemble seeks to mitigate the temporal blindness and computational constraints inherent in isolated modeling approaches, while keeping in mind the stringent operational guidelines set by the A²I Roadmap.

3. Data and Methodology

3.1. Dataset description and Split configuration

For the purposes of this research, the spacecraft Anomaly Detection Challenge (ESA-ADB) dataset is partitioned into two primary segments: a 14-year training set equipped with binary ground-truth labels and an unlabeled six-month test fragment for final evaluation. To ensure methodological rigor, the labeled training set was further divided into training and validation subsets using an approximate 80/20 training–validation split. Additionally, a Reduced Dataset Variant focusing on channels 41–46 was utilized during the prototyping and ablation stages; these channels were pre-identified by mission experts as a self-contained representation of the spacecraft's critical behavior.

3.2. Preprocessing and signal stability

Numerical stability and comparability across heterogeneous telemetry sensors, such as disparate voltage and temperature scales, were achieved through the application of Z-score normalization. To maintain the integrity of the evaluation and prevent data leakage, normalization parameters—specifically the mean (μ) and standard deviation (σ)—were computed exclusively on the training split and subsequently projected onto the validation and test sets. The preprocessing pipeline further involved replacing original timestamps with integer indices and aggregating channel-specific annotations into a single point-level binary label, where a time step is marked as anomalous if at least one monitored channel deviates from nominal behavior. Furthermore, the core execution script was modified to include a robust column-checking mechanism, ensuring that the LSTM inference pipeline remained stable even during the processing of unlabeled test streams, thus avoiding potential matrix size-incompatibility errors.

3.3. Baseline Statistical Detectors

Prior to the development of complex ensemble architectures, two classical statistical methods were implemented to serve as reference benchmarks. Global Statistical Thresholding was utilized to identify points deviating significantly from the 14-year mission mean: 1 shows what happens when setting the parameter `N` to 3 (meaning how many observations lie at least 3 standard deviations from the mean) while in 2 the parameter is set to 5; Local Statistical Thresholding employed sliding-window statistics to adapt to the non-stationary shifts inherent in long-term spacecraft operations. Preliminary analysis suggested that these methods produced significant overlap between nominal and anomalous score distributions. Due to this low discriminative power and high sensitivity to operational mode changes, they were used as performance baselines but excluded from the final ensemble to avoid introducing noise.

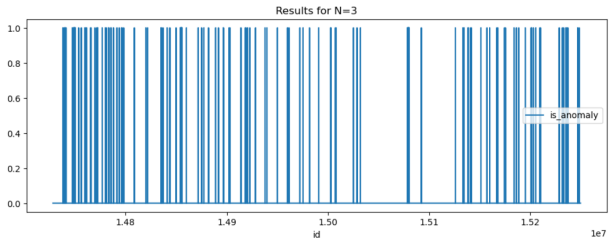

Figure 1 | Global STD with parameter N=3

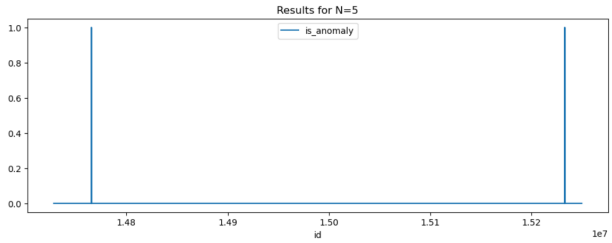

Figure 2 | Global STD with parameter N=5

4. The Algorithm

4.1. Hybrid ensemble framework

The proposed methodology implements a multi-stage hybrid ensemble designed to capture the diverse manifestations of spacecraft anomalies by integrating structural and temporal perspectives. The foundation of this ensemble rests on two complementary base detectors: Isolation Forest and Telemanom. The Isolation Forest algorithm was configured with 100 trees and a 10 % contamination factor to identify multivariate structural outliers through random partitioning of the feature space. It is specifically utilized to capture sudden spikes and structurally inconsistent states. To complement this structural perspective, a Telemanom-inspired temporal anomaly detector is employed. Rather than deploying the full original Telemanom LSTM architecture, a simplified prediction-error formulation is adopted in which future values are estimated from a sliding historical window using exponentially weighted moving averages. Anomaly scores are derived from instantaneous prediction errors together with a cumulative error component, enabling the detection of sequence-level deviations such as slow drifts and trajectory changes.

4.2. Meta-Learning fusion and explainability

To overcome the operational limitations of manual threshold tuning, which is often cited as a significant bottleneck in mission control, a supervised meta-learning layer based on XGBoost was implemented to fuse detector outputs. This fusion stage operates on a 16-dimensional engineered feature matrix that incorporates raw detector scores, rolling mean and standard deviation descriptors across 10, 30, and 50-step windows, and first-order delta and slope features to quantify the rate of change. The model utilizes a dynamic validation strategy to ensure robust learning under varying data availability. Model interpretability is supported through tree-based feature importance analysis provided by the XGBoost fusion model, enabling post-hoc assessment of the relative contribution of structural and temporal signals to anomaly likelihood.

4.3. Post processing and event formation

The final stage of the algorithm involves converting raw anomaly probabilities into discrete, operationally relevant events. This refinement process utilizes gap merging to connect fragmented detections separated by five or fewer time steps and minimum duration filtering to suppress isolated

"flickers" shorter than three time steps. These post-processing steps align the model output with the operational requirements of flight controllers, who prioritize persistent and actionable anomaly intervals over instantaneous fluctuations.

5. Evaluation

The evaluation of this ensemble solution demonstrates a high degree of synergy between spatial partitioning and temporal residual analysis, characterized by a strong Pearson correlation of $r = 0.933$ between the two base learners (iForest and Telemanom), as shown in 3. While both models show a dense concentration of points in the nominal cluster (near 0,0), the score distributions reveal that Telemanom produces a tighter distribution for normal data compared to the broader spread of iForest.

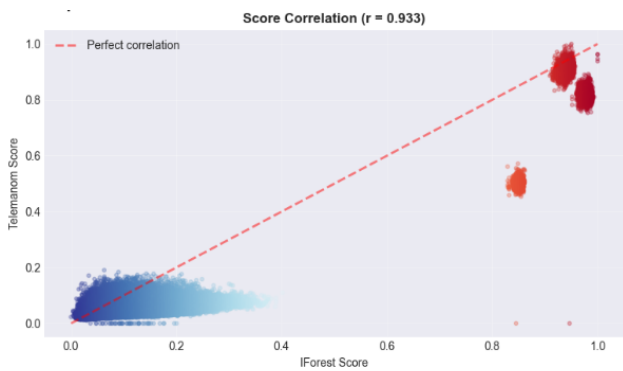

Figure 3 | Here the correlation between anomaly scores from the two models are depicted.

This statistical variance is further clarified by the agreement analysis at a 0.7 threshold, which shows that while 98.0% of the data is classified as "Both Normal" and 1.8% as "Both Anomaly," there remains a critical 0.2% of instances identified only by iForest (4). The XGBoost fusion layer effectively leverages these discrepancies, learning to weigh iForest's detection of unique spatial outliers against Telemanom's temporal residuals to minimize false positives while ensuring that subtle point anomalies, which represent the 996 instances missed by the temporal model, are accurately captured.

The evaluation of the XGBoost fusion layer demonstrates its effectiveness in synthesizing

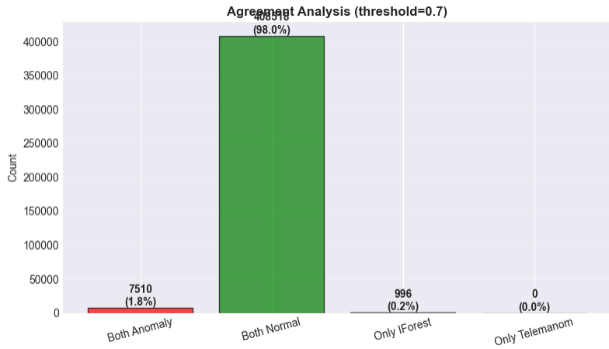

Figure 4 | Here the Agreement Analysis portrays that iForest has uniquely identified some instances

spatial and temporal features, as evidenced by a high model consensus $r = 0.933$ and superior feature prioritization. Feature importance analysis, as reported in 6, identifies the `iforest_score` as the primary driver of the ensemble, followed by rolling mean deviations `rolling_mean_dev_50` and `rolling_mean_dev_30`, which indicates that the XGBoost layer successfully learns to interpret raw anomaly scores within their local statistical context. While the individual base learners exhibit differing score distributions, Telemanom producing a tighter nominal range compared to the broader spread of iForest, the fusion layer reconciles these differences to achieve significant class separation. The "Score Separation (Sorted)" and "Prediction Probability Distribution" plots confirm this, showing a sharp bifurcation between normal and anomalous samples with very few instances falling near the 0.5 decision threshold. Quantitatively, the fusion model achieved a successful detection of 2,225 anomalies with a remarkably low false positive count of 81 (as shown in the confusion matrix), proving that the XGBoost layer effectively mitigates the noise inherent in individual detectors while maintaining high sensitivity to true system deviations.

To refine the final system output, a post-processing stage involving gap merging and length filtering was implemented to improve the temporal coherence of the detected anomalies. As illustrated in 8, the initial model predictions exhibited high-frequency oscillations and fragmented anomaly windows, which can lead to

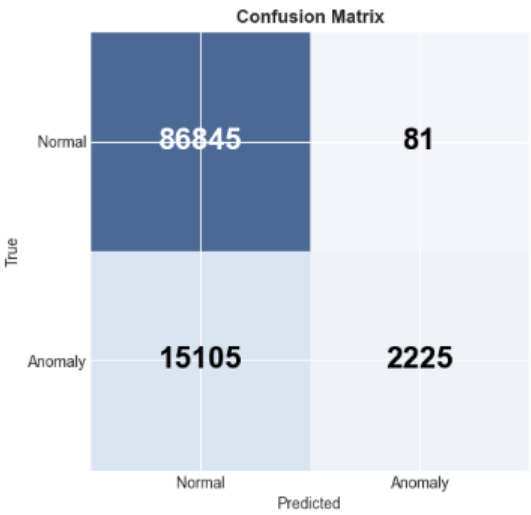

Figure 5 | Confusion Matrix showing the results of the XGBoost layer

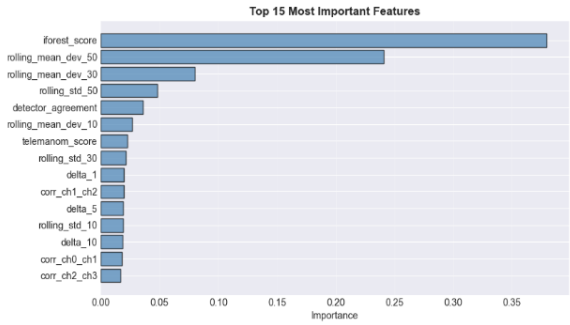

Figure 6 | The feature importance plot shows a significance importance for the iForest score and the rolling windows

excessive alerting in operational environments. By applying a temporal smoothing logic that merges proximal detections and filters out transient spikes below a specific duration threshold, the system produced a more stable and continuous representation of the anomalous events. This refinement ensures that the ensemble does not just identify isolated statistical outliers, but rather captures sustained system deviations that are more indicative of genuine failures, thereby providing a cleaner, more actionable signal for end-users while maintaining the high detection sensitivity established by the XGBoost fusion layer.

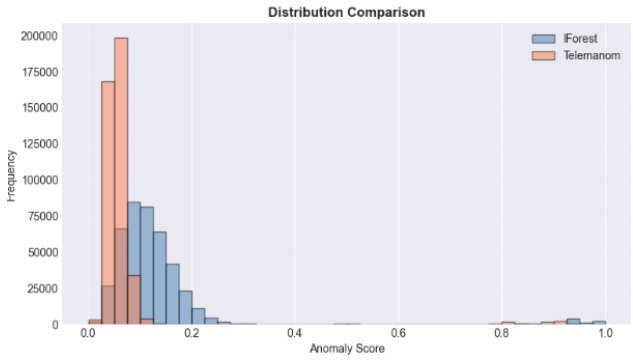

Figure 7 | Distribution Comparison among the anomaly scores of the two detectors (iForest and Telemanom)

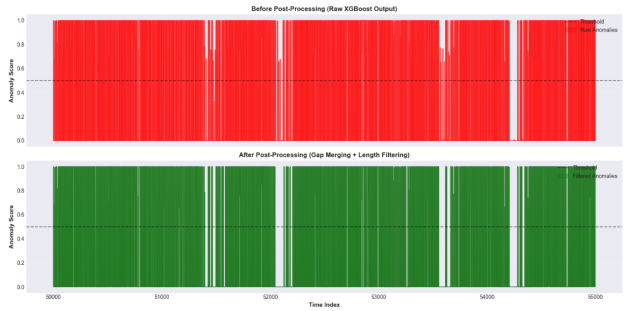

Figure 8 | Comparison of the XGBoost output before and after the Post-processing

6. Conclusion

When dealing with telemetry data in the space sector, there is the need to be careful in adopting such *blackbox* technologies, since the understanding of the output and the features' influence is vital. Identified Future enhancements could be introduced by addressing the reduction in False Positives which represent a threat to this sector (by maximizing the `f0.5 score`, as a more suitable evaluation metric for this problem); performing an efficient hyperparameter tuning, which hasn't been done due to lack of computational power; and a better identification of the threshold in the postprocessing phase.

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